

Mahidol University
Faculty of Information
and Communication Technology

Package Management, Finding Linux Documents, and Introduction to Processes

ITCS 254 | Lecture 5

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Agenda

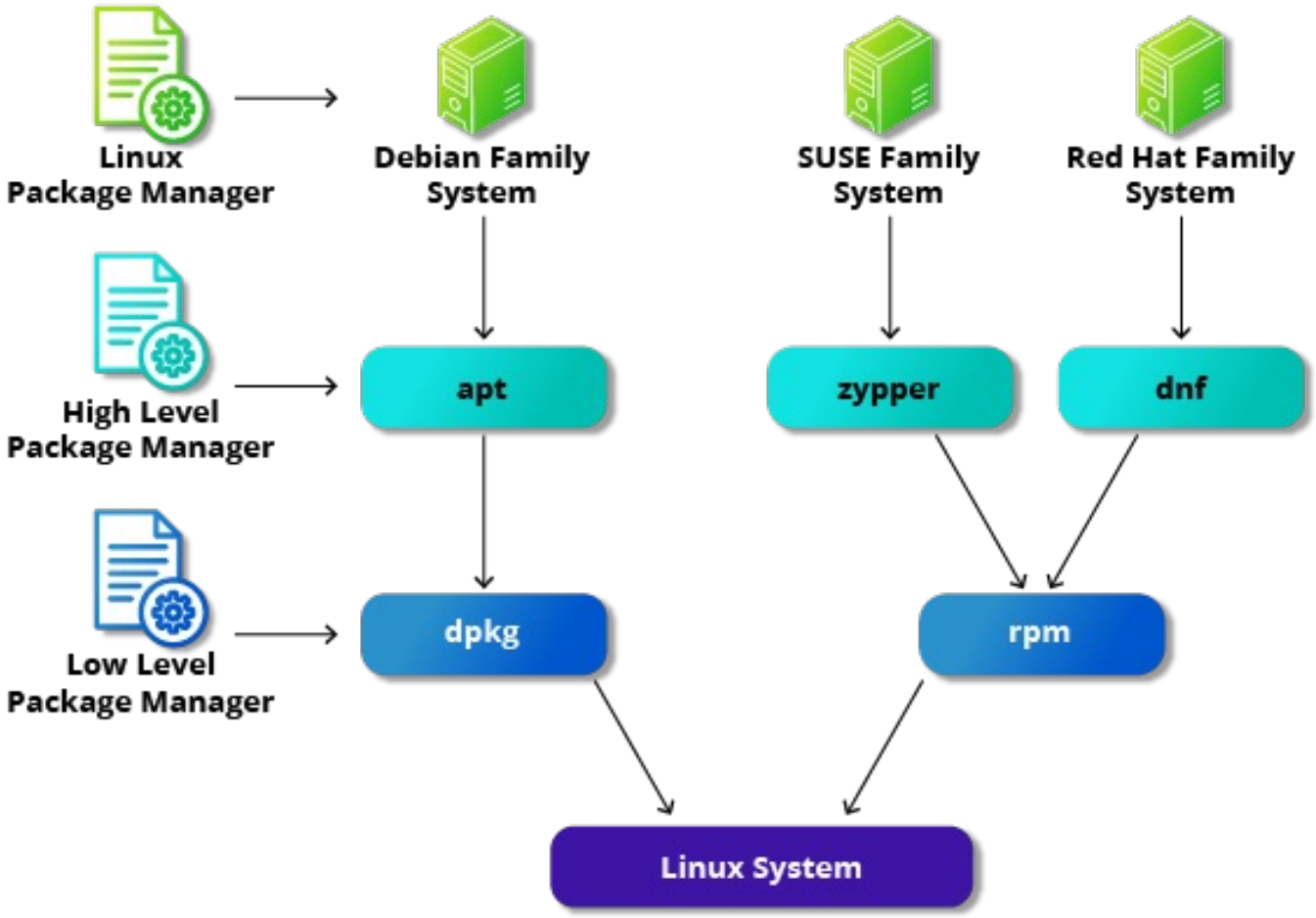
- Installing and Updating Software
- Documentation Sources
- The `man` Pages
- GNU `info`
- The `--help` Options and Help Command
- Other Documentation Sources
- Introduction to Processes

Package Management Systems on Linux

- The core parts of a Linux distribution and most of its add-on software are installed via the Package Management System.
- Each package contains the files and other instructions needed to make one software component work well and cooperate with the other components that comprise the entire system.
- Packages can depend on each other.
- There are two broad families of package managers widely deployed: those based on Debian and those which use RPM as their low-level package manager.
  - The two systems are incompatible but, broadly speaking, provide the same essential features and satisfy the same needs.
  - There are some other systems used by more specialized Linux distributions.

Package Managers

- A low-level tool takes care of the details of unpacking individual packages, running scripts, getting the software installed correctly
- A high-level tool works with groups of packages, downloads packages from the vendor, and figures out dependencies.



Working With Different Package Management Systems

- The Advanced Packaging Tool (`apt`) is the underlying package management system that manages software on Debian-based systems.
    - While it forms the backend for graphical package managers, such as the Ubuntu Software Center and synaptic, its native user interface is at the command line, with programs that include `apt` (or `apt-get`) and `apt-cache`.
- `dnf` is the open source command-line package-management utility for the RPM-compatible Linux systems that belong to the Red Hat family.
- `zypper` is the package management system for the SUSE/openSUSE family and is also based on RPM.
    - zypper also allows you to manage repositories from the command line.
    - zypper is fairly straightforward to use and closely resembles dnf.

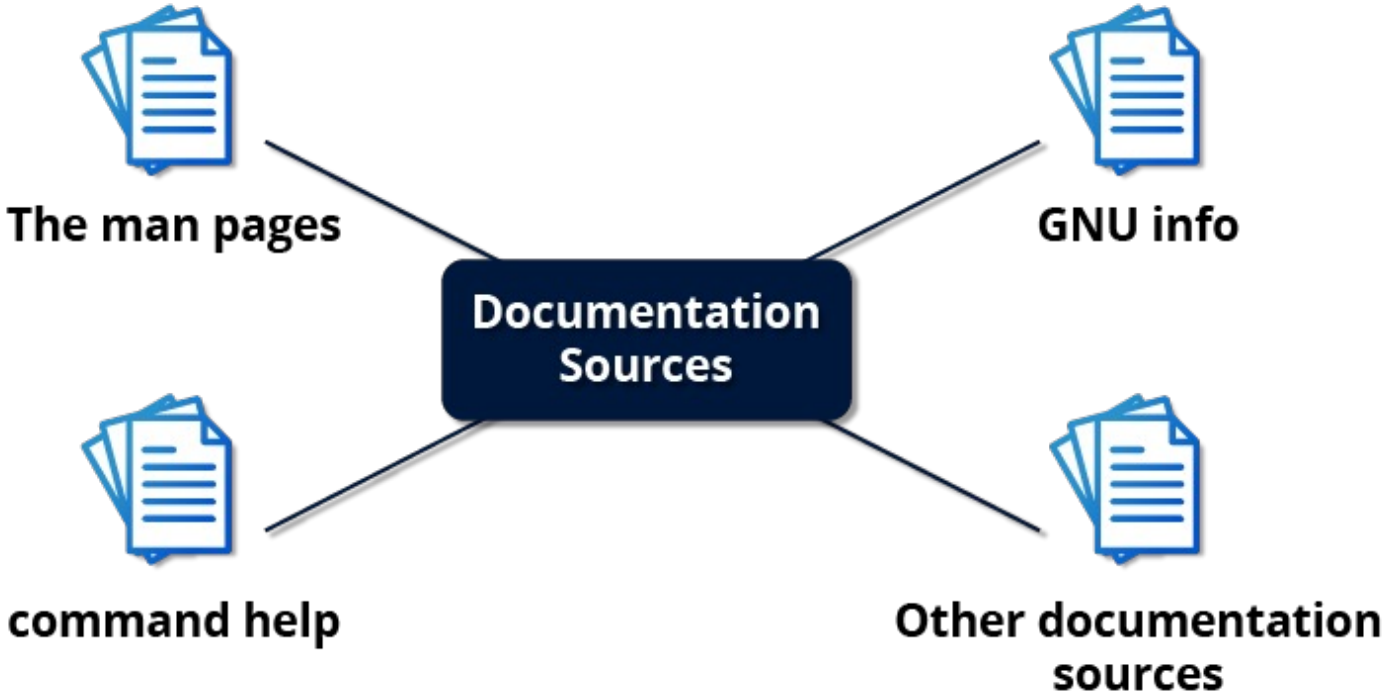
Basic Packaging Commands

Operation	RPM	DEB
Install package	`rpm -i foo.rpm`	`dpkg --install foo.deb`
Install package, dependencies	`dnf install foo`	`apt install foo`
Remove package	`rpm -e foo.rpm`	`dpkg --remove foo.deb`
Remove package, dependencies	`dnf remove foo`	`apt autoremove foo`
Update package	`rpm -U foo.rpm`	`dpkg --install foo.deb`
Update package, dependencies	`dnf update foo`	`apt install foo`
Update entire system	`dnf update`	`apt dist-upgrade`
Show all installed packages	`rpm -qa` or `dnf list installed`	`dpkg --list`
Get information on package	`rpm -qil foo`	`dpkg --listfiles foo`
Show packages named foo	`dnf list "foo"`	`apt-cache search foo`
Show all available packages	`dnf list`	`apt-cache dumpvail foo`
What package is `file` part of?	`rpm -qf file`	`dpkg --search file`

Basic `apt` Commands

• `sudo apt-get update`	Gets the latest package versions.
• `sudo apt-cache search <keyword>`	Finds a package by keyword.
• `sudo apt-cache show <pkgname>`	Displays information about a package.
• `sudo apt-get install <pkgname>`	Installs the *pkgname* package.
• `sudo apt-get upgrade`	Updates installed packages if upgrade is ready.
• `sudo apt-cache depend <pkgname>`	Shows what packages <pkgname> depends on.
• `sudo apt-get remove <pkgname>`	Uninstall a package

Linux Documentation Sources

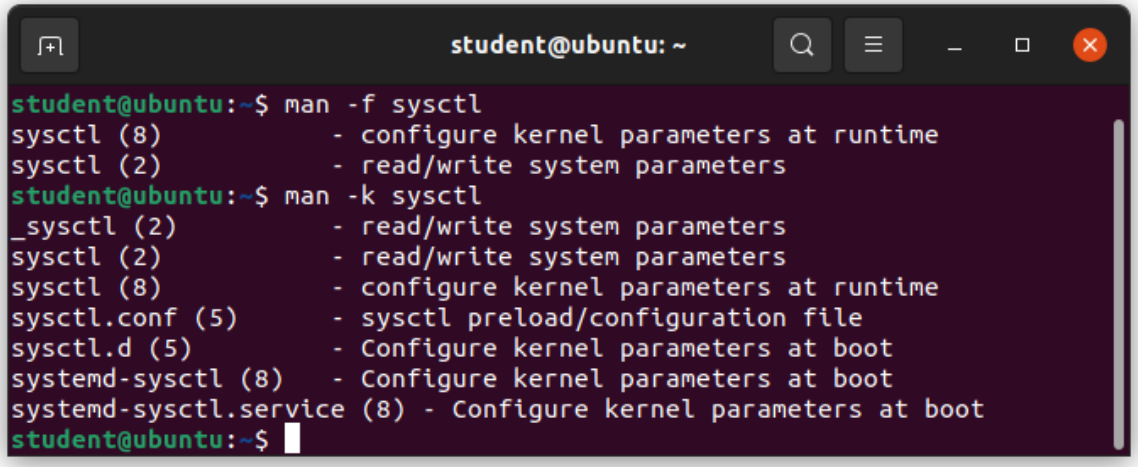


The `man` Pages

- They provide in-depth documentation about many programs and utilities, as well as other topics, including configuration files and programming APIs for system calls, library routines, and the kernel.
- The man pages infrastructure was first introduced in the early UNIX versions, at the beginning of the 1970s.
- `man` pages are often converted to other formats, such as PDF documents and web pages.
- http://man7.org/linux/man-pages/
- Other sources of documentation include published books and many Internet sites.

man

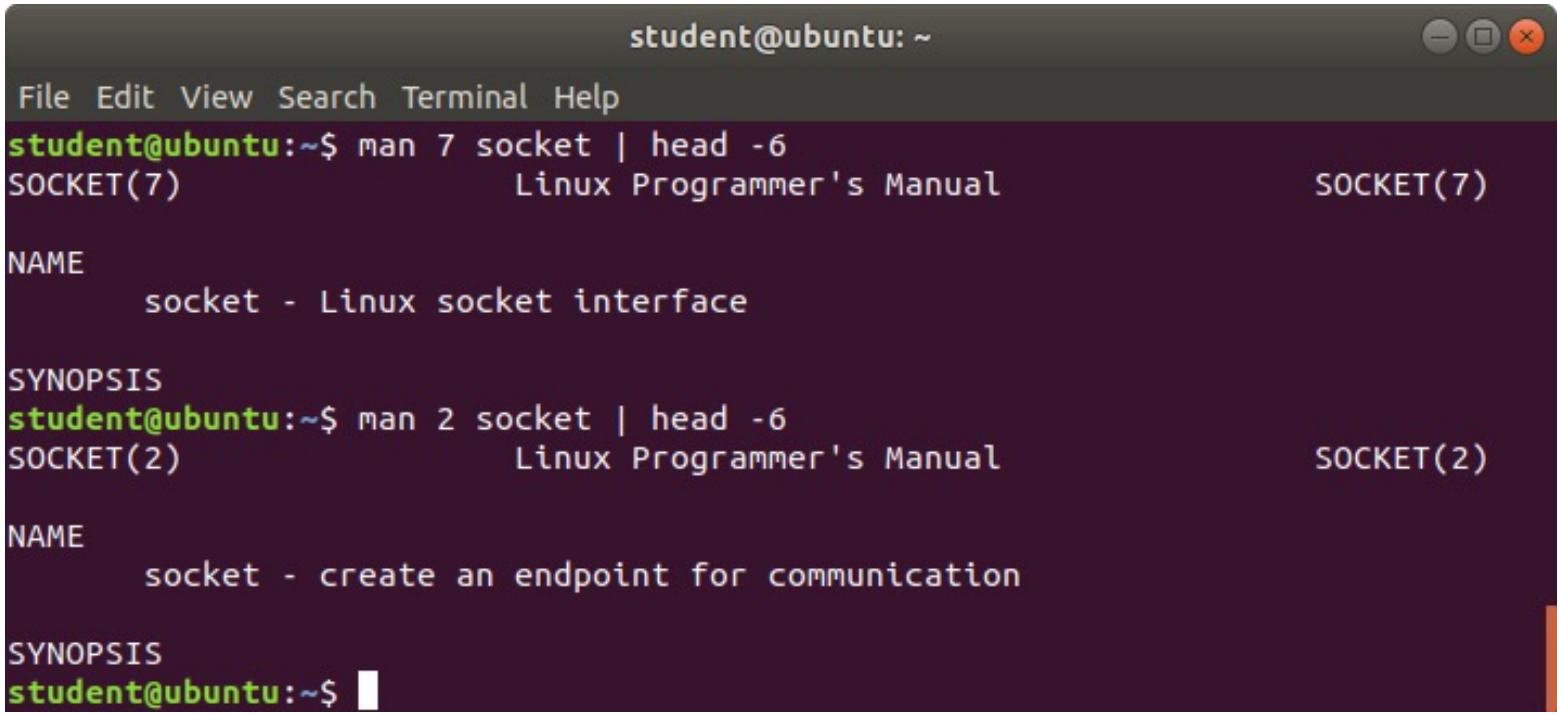
- The `man` program searches, formats, and displays the information contained in the man page system.
- A given topic may have multiple pages associated with it and there is a default order determining which one is displayed when no options or section number is specified.
- `man -f` generates the same result as typing `whatis`.
- `man -k` generates the same result as typing `apropos`.
- The default order is specified in `/etc/man_db.conf` and is roughly (but not exactly) in ascending numerical order by section.



Manual Chapters [1/2]

- The man pages are divided into chapters numbered 1 through 9.
- In some cases, a letter is appended to the chapter number to identify a specific topic.
  - For example, many pages describing part of the X Window API are in chapter 3X.
- The chapter number can be used to force man to display the page from a particular chapter.
- It is common to have multiple pages across multiple chapters with the same name, especially for names of library functions or system calls.
- `man -a <name>`
  - `man` will display all pages with the given name in all chapters, one after the other.

Manual Chapters [2/2]



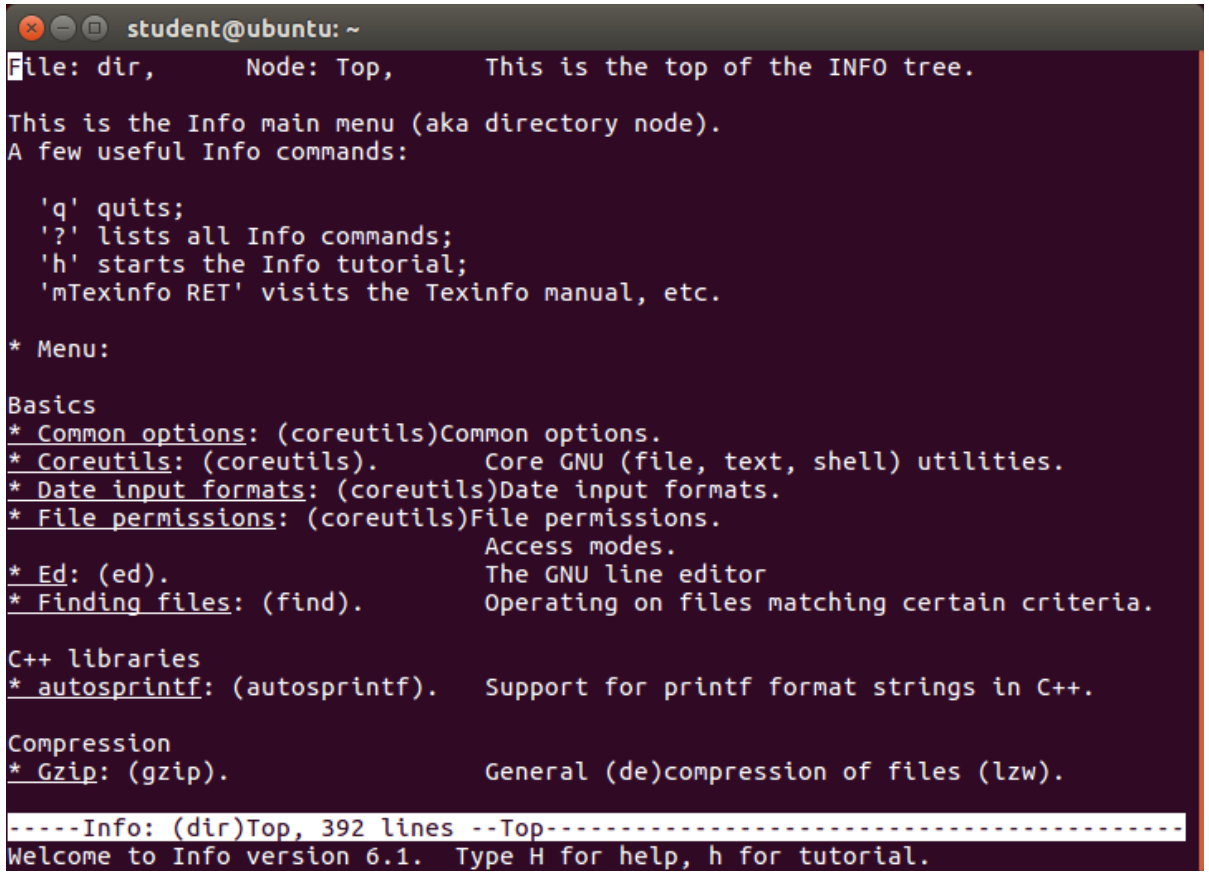
The GNU Info System



- It prefers as an alternative to `man`.
- The Info System is basically free-form and supports linked subsections.
- Functionally, info resembles man in many ways. However, topics are connected using links (even though its design predates the World Wide Web).
- Information can be viewed through either a command line interface, a graphical help utility, printed or viewed online.
- While the info interface may seem to be rather outdated compared to modern help systems (even `man`), it often is the only easy source to get more complete information.

Using info

- You can view help for a particular topic by typing `info <topic-name>`.
- Some useful keys are: `q` to quit, `h` for help, and Enter to select a menu item.



`info` Page Structure

- The topic which you view in an info page is called a node.
- The table lists the basic keystrokes for moving between nodes.
- Nodes are essentially sections and subsections in the documentation.
- You can move between nodes or view each node sequentially.
- Each node may contain menus and linked subtopics, or items.
- Items function like browser links and are identified by an asterisk (`*`) at the beginning of the item name.
- Named items (outside a menu) are identified with double-colons (`::`) at the end of the item name.
- Items can refer to other nodes within the file or to other files.

Key	Function
`n`	Go to the next node.
`p`	Go to the previous node
`u`	Move one node up in the index

The `--help` Option

- Most commands have an available short description which can be viewed using the `--help` or the `-h` option along with the command or application.
- The `--help` option is useful as a quick reference and it displays information faster than the `man` or `info` pages.

```
File  Edit  View  Search  Terminal  Help
C:/tmp>man --help
Usage: man [OPTION...] [SECTION] PAGE...

  -C, --config-file=FILE     use this user configuration file
  -d, --debug                emit debugging messages
  -D, --default              reset all options to their default values
      --warnings[=WARNINGS]  enable warnings from groff

 Main modes of operation:
  -f, --whatis               equivalent to whatis
  -k, --apropos              equivalent to apropos
  -K, --global-apropos       search for text in all pages
  -l, --local-file           interpret PAGE argument(s) as local filename(s)
  -w, --where, --path, --location
                             print physical location of man page(s)
  -W, --where-cat, --location-cat
                             print physical location of cat file(s)

  -c, --catman               used by catman to reformat out of date cat pages
  -R, --recode=ENCODING      output source page encoded in ENCODING

 Finding manual pages:
  -L, --locale=LOCALE        define the locale for this particular man search
  -m, --systems=SYSTEM       use manual pages from other systems
  -M, --manpath=PATH         set search path for manual pages to PATH

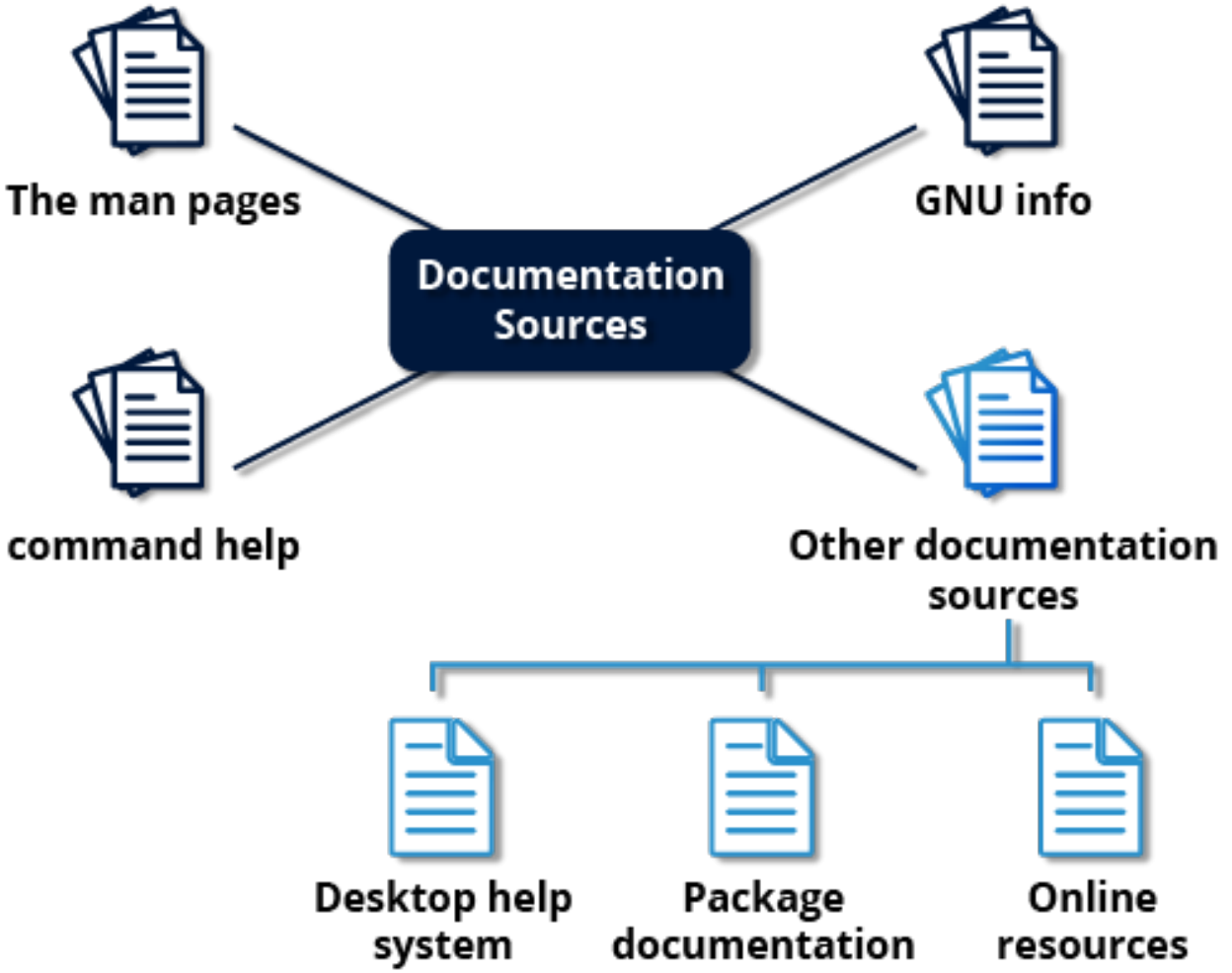
  -S, -s, --sections=LIST    use colon separated section list

  -e, --extension=EXTENSION  limit search to extension type EXTENSION

  -i, --ignore-case          look for pages case-insensitively (default)
  -I, --match-case           look for pages case-sensitively

      --regex                show all pages matching regex
/tmp/manhelp
```

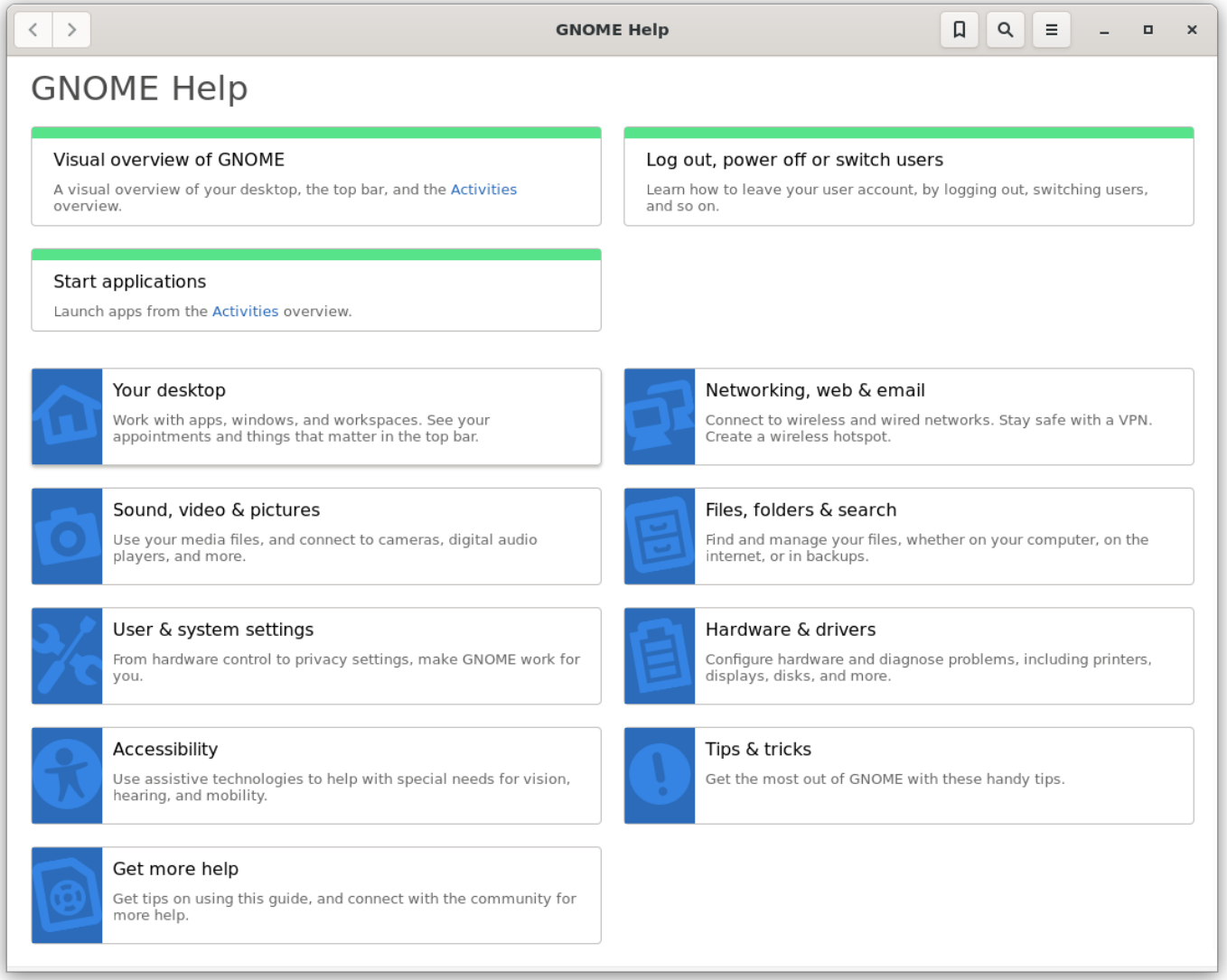
Other Documentation Sources



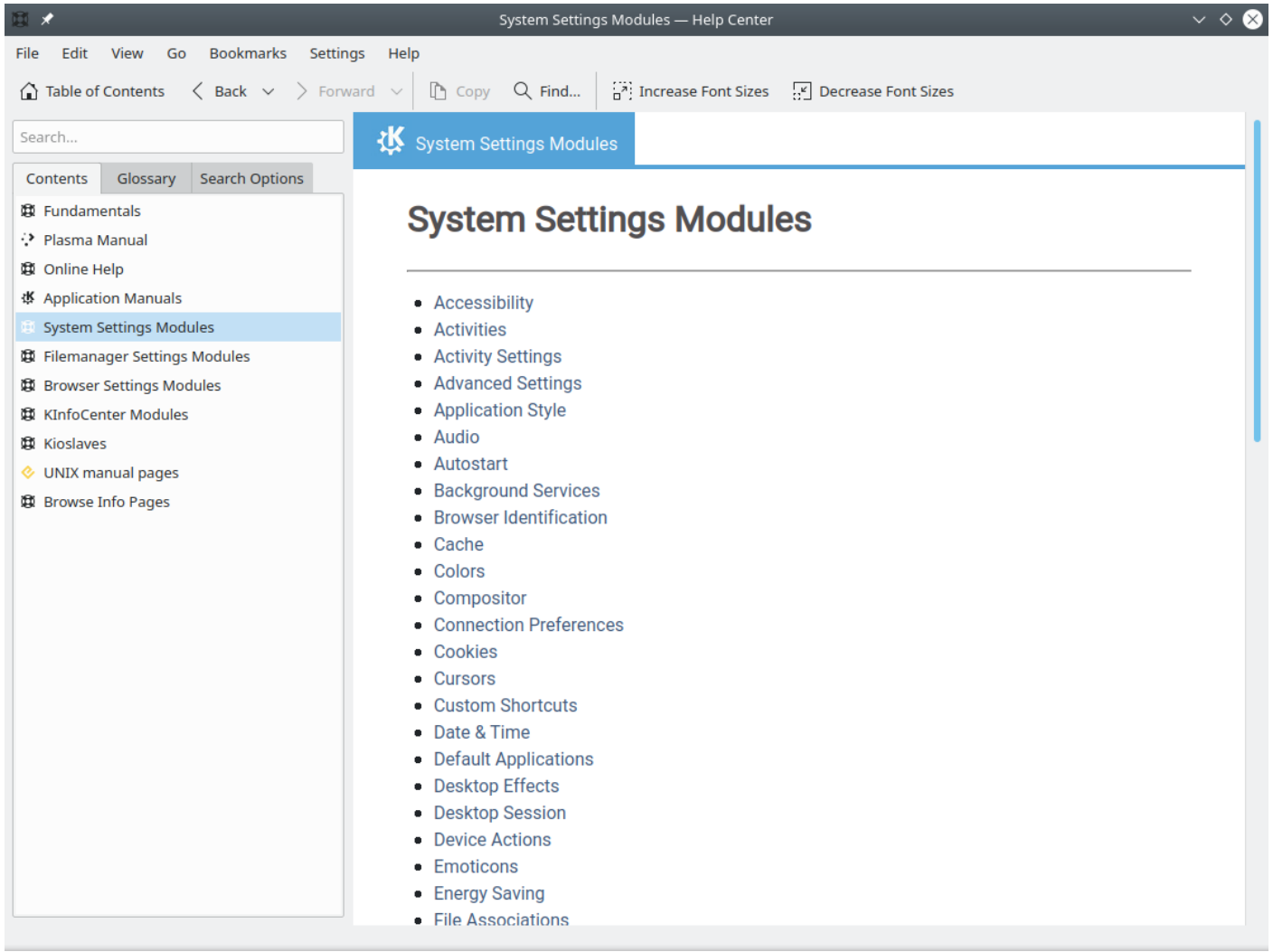
Graphical Help Systems

- All Linux desktop systems have a graphical help application.
- This application is usually displayed as a question-mark icon or an image of a ship's life preserver, and it can also always be found within the menu system.
- These programs usually contain custom help for the desktop itself and some of its applications, and they will sometimes also include graphically-rendered `info` and `man` pages.
- You can also start the graphical help system from a terminal window or command prompt by
  - GNOME: `gnome-help` or `yelp`
  - KDE: `khelpcenter`

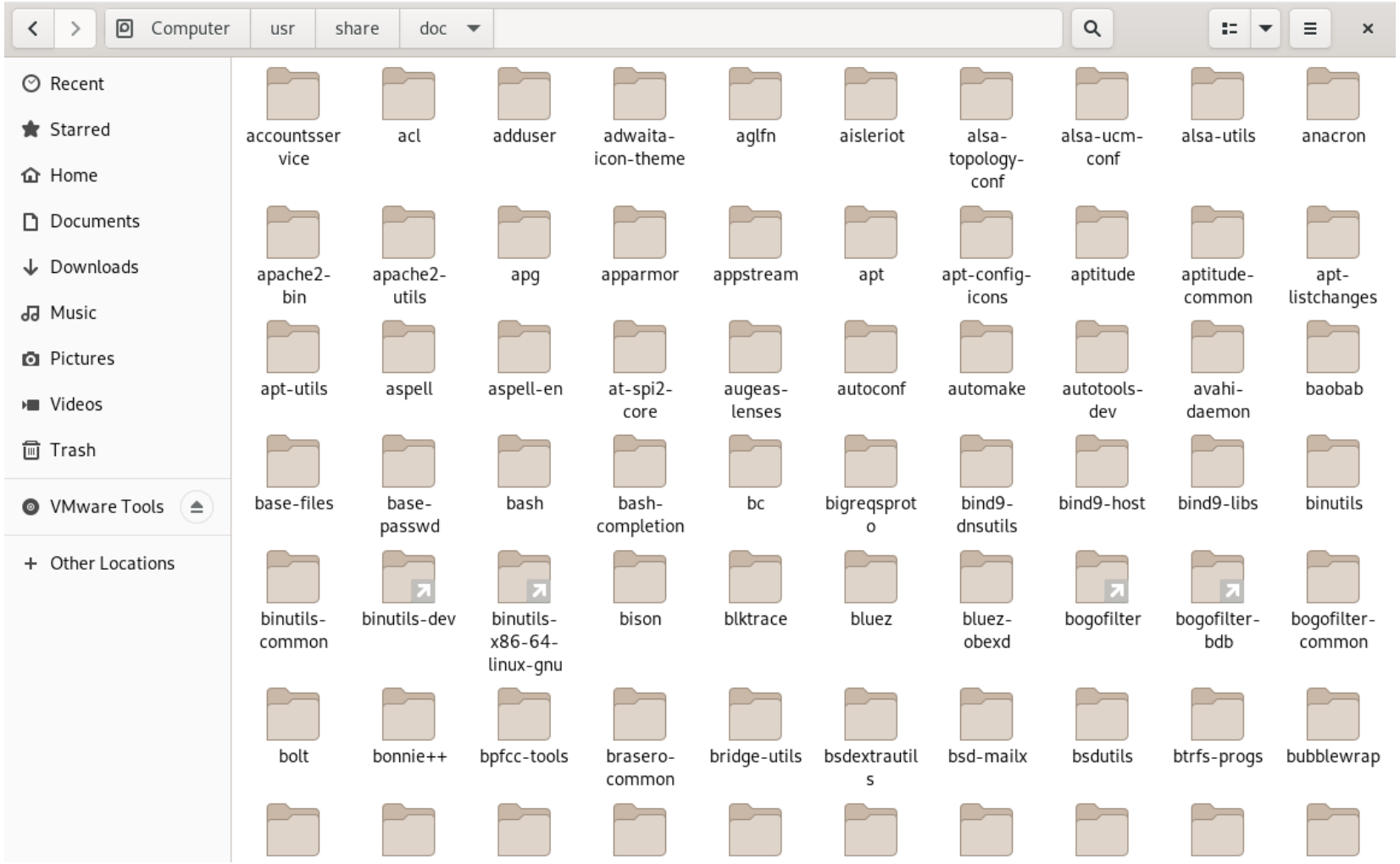
GNOME Help


GNOME Help
GNOME Help
Visual overview of GNOME
A visual overview of your desktop, the top bar, and the Activities overview.
Log out, power off or switch users
Learn how to leave your user account, by logging out, switching users, and so on.
Start applications
Launch apps from the Activities overview.
Your desktop
Work with apps, windows, and workspaces. See your appointments and things that matter in the top bar.
Networking, web & email
Connect to wireless and wired networks. Stay safe with a VPN. Create a wireless hotspot.
Sound, video & pictures
Use your media files, and connect to cameras, digital audio players, and more.
Files, folders & search
Find and manage your files, whether on your computer, on the internet, or in backups.
User & system settings
From hardware control to privacy settings, make GNOME work for you.
Hardware & drivers
Configure hardware and diagnose problems, including printers, displays, disks, and more.
Accessibility
Use assistive technologies to help with special needs for vision, hearing, and mobility.
Tips & tricks
Get the most out of GNOME with these handy tips.
Get more help
Get tips on using this guide, and connect with the community for more help.

KDE Help


System Settings Modules — Help Center
File Edit View Go Bookmarks Settings Help
Table of Contents Back Forward Copy Find... Increase Font Sizes Decrease Font Sizes
Search...
Contents Glossary Search Options
Fundamentals
Plasma Manual
Online Help
Application Manuals
System Settings Modules
Filemanager Settings Modules
Browser Settings Modules
KInfoCenter Modules
Kioslaves
UNIX manual pages
Browse Info Pages
System Settings Modules
System Settings Modules
Accessibility
Activities
Activity Settings
Advanced Settings
Application Style
Audio
Autostart
Background Services
Browser Identification
Cache
Colors
Compositor
Connection Preferences
Cookies
Cursors
Custom Shortcuts
Date & Time
Default Applications
Desktop Effects
Desktop Session
Device Actions
Emoticons
Energy Saving
File Associations

Package Documentation

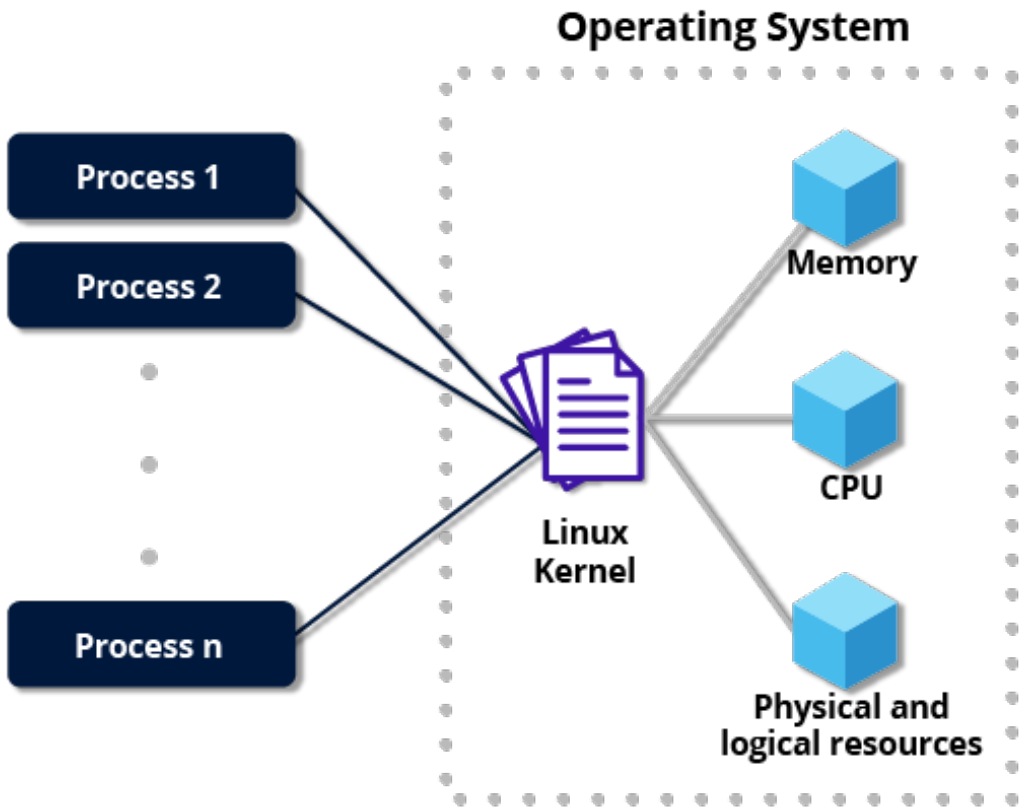


Online Resources

- The Linux Command Line by William Shotts.
- Ubuntu Documentation
- CentOS Documentation
- openSUSE Documentation
- Gentoo Documentation
- Fedora Documentation

What is a Process?

- A process is simply an instance of one or more related tasks (threads) executing on your computer. It is not the same as a program or a command.
- A single command may actually start several processes simultaneously.
- Some processes are independent of each other and others are related.
- A failure of one process may or may not affect the others running on the system.
- A terminal window (one kind of command shell) is a process that runs as long as needed.
    - It allows users to execute programs and access resources in an interactive environment.
    - You can also run programs in the background, which means they become detached from the shell.

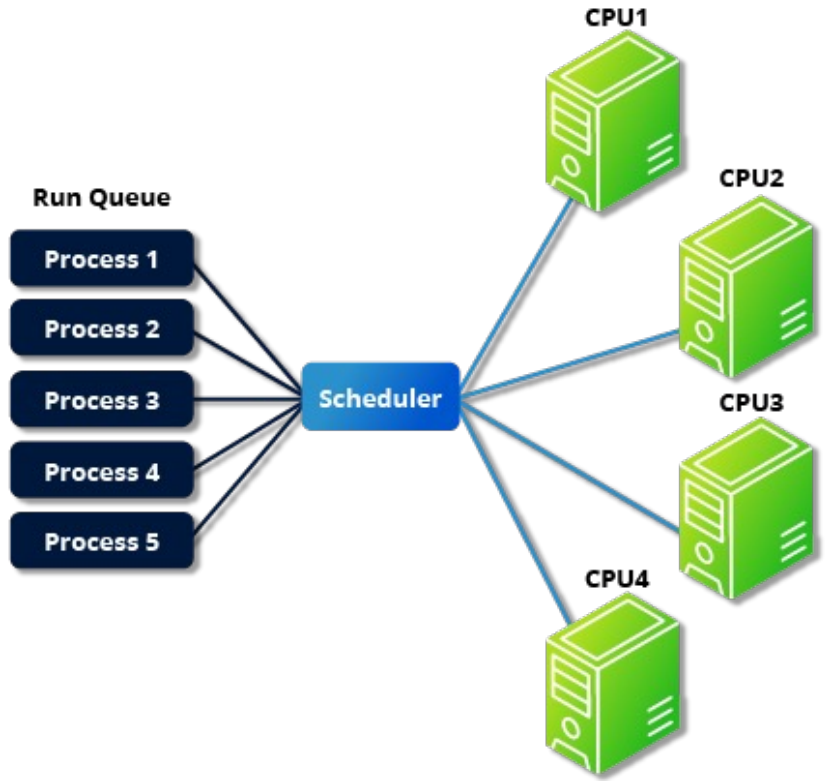


Process Types

Process type	Description	Examples
Interactive processes	Need to be started by a user, either at a command line or through a graphical interface such as an icon or a menu selection.	bash, firefox, top, Slack, Libreoffice
Batch processes	Automatic processes which are scheduled from and then disconnected from the terminal. These tasks are queued and work on a FIFO basis.	updatedb, ldconfig
Daemons	Server processes that run continuously. Many are launched during system startup and then wait for a user or system request indicating that their service is required.	httpd, sshd, libvirtd, cupsd
Threads	Lightweight processes. These are tasks that run under the umbrella of a main process, sharing memory and other resources, but are scheduled and run by the system on an individual basis. An individual thread can end without terminating the whole process and a process can create new threads at any time. Many non-trivial programs are multi-threaded.	dconf-service, gnome-terminal-server
Kernel threads	Kernel tasks that users neither start nor terminate and have little control over. These may perform actions like moving a thread from one CPU to another, or making sure input/output operations to disk are completed.	kthreadd, migration, ksoftirqd

Process Scheduling and States [1/2]

- Scheduler: A critical kernel function constantly shifts processes on and off the CPU, sharing time according to relative priority, how much time is needed and how much has already been granted to a task.
- Running state
  - A process is either currently executing instructions on a CPU or is waiting to be granted a share of time (a time slice) so it can execute.
  - All processes in this state reside on what is called a run queue and on a computer with multiple CPUs, or cores, there is a run queue on each.



Process Scheduling and States [2/2]

- Sleep state
  - Generally, when processes are waiting for something to happen before they can resume, perhaps for the user to type something.
  - In this condition, a process is said to be sitting in a wait queue.
- Zombie state
  - There are less frequent process states, especially when a process is terminating.
  - Sometimes, a child process completes, but its parent process has not asked about its state.
  - Amusingly, such a process is said to be in a zombie state; it is not really alive, but still shows up in the system's list of processes.